1. Write a C program to simulate dining philosopher problem.

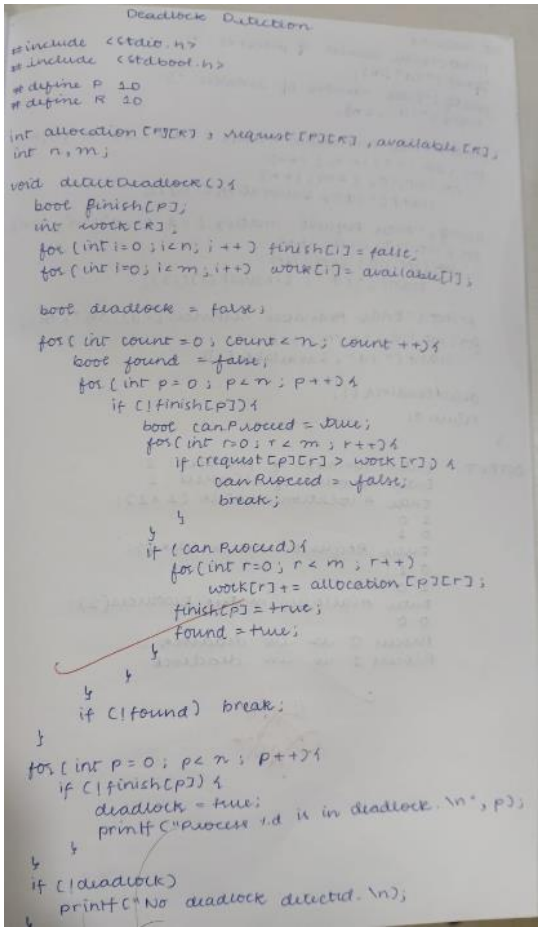
```
#include <stdio.h>
#include <stdlib.h>
#include <pthread.h>
#include <semaphore.h>
#include <unistd.h>

#define N 5

sem_t chopstick[N];
pthread_mutex_t mutex;

void *philosopher(void *num) {
   int id = *(int *)num;

   while (1) {
      pthread_mutex_lock(&mutex);
      printf("Philosopher %d status: THINKING\n", id);
      pthread_mutex_unlock(&mutex);

      sleep(1);

      pthread_mutex_lock(&mutex);
      printf("Philosopher %d status: WAITING\n", id);
      pthread_mutex_unlock(&mutex);

      sem_wait(&chopstick[id]);
      sem_wait(&chopstick[(id + 1) % N]);

      pthread_mutex_lock(&mutex);
```

```
        printf("Philosopher %d status: EATING\n", id);
        pthread_mutex_unlock(&mutex);

        sleep(2);

        sem_post(&chopstick[id]);
        sem_post(&chopstick[(id + 1) % N]);
    }
}

int main() {
    pthread_t tid[N];
    int i, ids[N];

    pthread_mutex_init(&mutex, NULL);

    for (i = 0; i < N; i++)
        sem_init(&chopstick[i], 0, 1);

    for (i = 0; i < N; i++) {
        ids[i] = i;
        pthread_create(&tid[i], NULL, philosopher, &ids[i]);
    }

    for (i = 0; i < N; i++)
        pthread_join(tid[i], NULL);

    return 0;
}
```

Output:

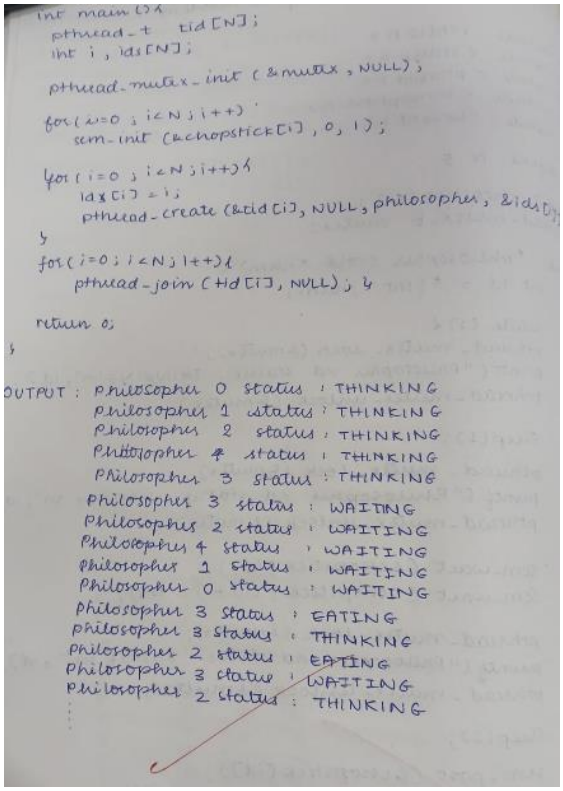
```
C:\Users\student\Documents\
Philosopher 0 status: THINKING
Philosopher 1 status: THINKING
Philosopher 2 status: THINKING
Philosopher 3 status: THINKING
Philosopher 4 status: THINKING
Philosopher 4 status: WAITING
Philosopher 4 status: EATING
Philosopher 1 status: WAITING
Philosopher 1 status: EATING
Philosopher 2 status: WAITING
Philosopher 3 status: WAITING
Philosopher 0 status: WAITING
Philosopher 1 status: THINKING
Philosopher 3 status: EATING
Philosopher 4 status: THINKING
Philosopher 0 status: EATING
Philosopher 4 status: WAITING
Philosopher 1 status: WAITING
```

Observation:



Deadlock Detection

```
#include <stdio.h>
#include <stdbool.h>
#define P 10
#define R 10

int allocation[P][R], request[P][R], available[R];
int n, m;

void detectDeadlock(){
  bool finish[P];
  int work[R];
  for(int i=0;i<n;i++) finish[i] = false;
  for(int i=0;i<m;i++) work[i] = available[i];

  bool deadlock = false;

  for(int count=0; count<n; count++){
    bool found = false;
    for(int p=0; p<n; p++){
      if(!finish[p]){
        bool canProceed = true;
        for(int r=0; r<m; r++){
          if(request[p][r] > work[r]){
            canProceed = false;
            break;
          }
        }
        if(canProceed){
          for(int r=0; r<m; r++)
            work[r] += allocation[p][r];
          finish[p] = true;
          found = true;
        }
      }
    }
    if(!found) break;
  }
  for(int p=0; p<n; p++)
    if(!finish[p]){
      deadlock = true;
      printf("Process %d is in deadlock.\n", p);
    }
  if(!deadlock)
    printf("No deadlock detected.\n");
}
```



```
int main()
{
    pthread_t tid[N];
    int i, ids[N];

    pthread_mutex_init(&mutex, NULL);

    for(i=0; i<N; i++)
        sem_init(&chopstick[i], 0, 1);

    for(i=0; i<N; i++){
        ids[i] = i;
        pthread_create(&tid[i], NULL, philosopher, &ids[i]);
    }
    for(i=0; i<N; i++){
        pthread_join(tid[i], NULL); }

    return 0;
}
```

OUTPUT :

```
Philosopher 0 status : THINKING
Philosopher 1 status : THINKING
Philosopher 2 status : THINKING
Philosopher 4 status : THINKING
Philosopher 3 status : THINKING
Philosopher 3 status : WAITING
Philosopher 2 status : WAITING
Philosopher 4 status : WAITING
Philosopher 1 status : WAITING
Philosopher 0 status : WAITING
Philosopher 3 status : EATING
Philosopher 3 status : THINKING
Philosopher 2 status : EATING
Philosopher 3 status : WAITING
Philosopher 2 status : THINKING
```

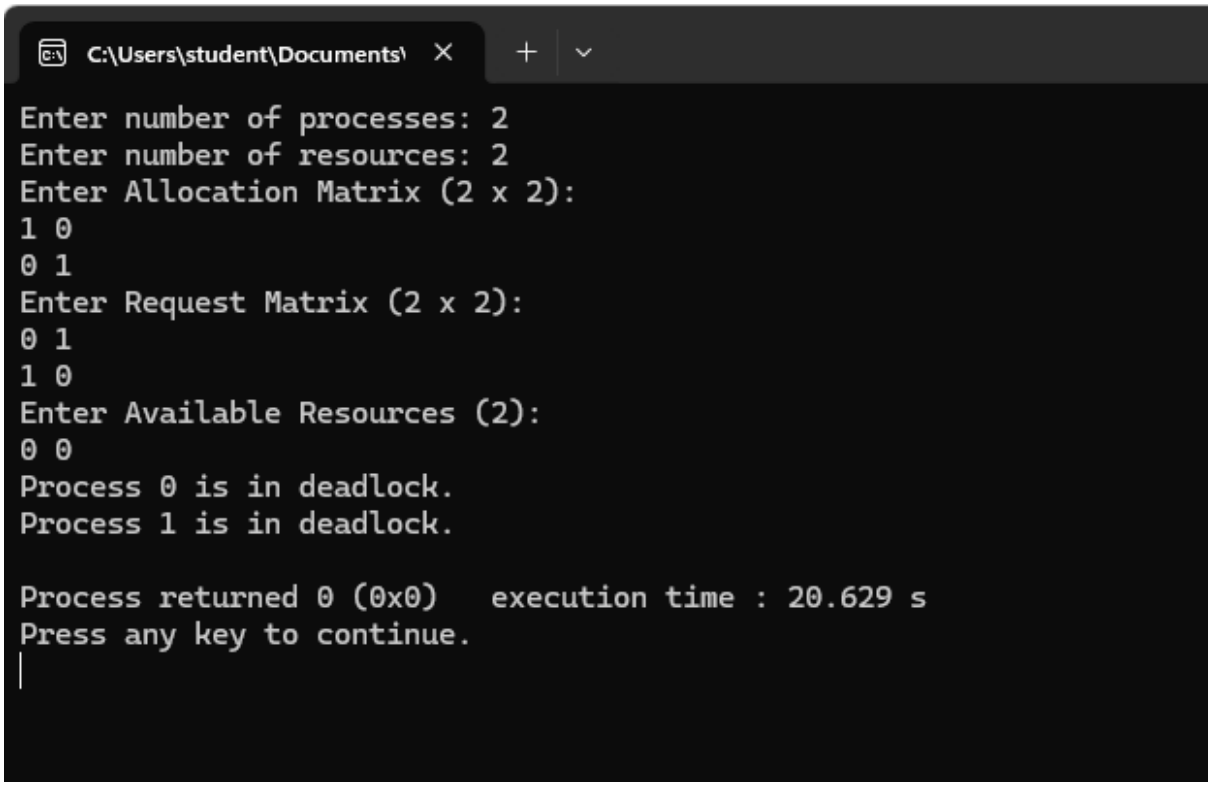
2. Write a C program to simulate deadlock detection algorithm. Use the examples solved in the class to verify your output.

```
#include <stdio.h>
#include <stdbool.h>

#define P 10
#define R 10

int allocation[P][R], request[P][R], available[R];
int n, m;

void detectDeadlock() {
   bool finish[P];
   int work[R];
   for (int i = 0; i < n; i++) finish[i] = false;
   for (int i = 0; i < m; i++) work[i] = available[i];

   bool deadlock = false;

   for (int count = 0; count < n; count++) {
      bool found = false;
      for (int p = 0; p < n; p++) {
         if (!finish[p]) {
            bool canProceed = true;
            for (int r = 0; r < m; r++) {
               if (request[p][r] > work[r]) {
                  canProceed = false;
                  break;
               }
            }
            if (canProceed) {
```

```
                for (int r = 0; r < m; r++)
                    work[r] += allocation[p][r];
                finish[p] = true;
                found = true;
            }
        }
    }
    if (!found) break;
  }

  for (int p = 0; p < n; p++) {
    if (!finish[p]) {
      deadlock = true;
      printf("Process %d is in deadlock.\n", p);
    }
  }

  if (!deadlock)
    printf("No deadlock detected.\n");
}

int main() {
  printf("Enter number of processes: ");
  scanf("%d", &n);
  printf("Enter number of resources: ");
  scanf("%d", &m);

  printf("Enter Allocation Matrix (%d x %d):\n", n, m);
  for (int i = 0; i < n; i++)
    for (int j = 0; j < m; j++)
      scanf("%d", &allocation[i][j]);
```

```
    printf("Enter Request Matrix (%d x %d):\n", n, m);
    for (int i = 0; i < n; i++)
        for (int j = 0; j < m; j++)
            scanf("%d", &request[i][j]);

    printf("Enter Available Resources (%d):\n", m);
    for (int i = 0; i < m; i++)
        scanf("%d", &available[i]);

    detectDeadlock();
    return 0;
}
```

Output:



Observation:

11/5/26 Dining Philosopher Problem

```
#include <stdio.h>
#include <stdlib.h>
#include <pthread.h>
#include <semaphore.h>
#include <unistd.h>

#define N 5

sem_t chopstick[N];
pthread_mutex_t mutex;

void *philosopher(void *num) {
    int id = *(int *)num;

    while(1) {
    pthread_mutex_lock(&mutex);
    printf("Philosopher %d status : THINKING\n", id);
    pthread_mutex_unlock(&mutex);

    sleep(1);
    pthread_mutex_lock(&mutex);
    printf("Philosopher %d status : WAITING \n", id);
    pthread_mutex_unlock(&mutex);

    sem_wait(&chopstick[id]);
    sem_wait(&chopstick[(id+1) % N]);

    pthread_mutex_lock(&mutex);
    printf("Philosopher %d status : EATING \n", id);
    pthread_mutex_unlock(&mutex);

    sleep(2);
    sem_post(&chopstick[id]);
    sem_post(&chopstick[(id+1) % N]);
    }
}
```

```
int main() {
    printf("Enter number of processes: ");
    scanf("%d", &n);
    printf("Enter number of resources: ");
    scanf("%d", &m);

    printf("Enter Allocation matrix (%d x %d):\n", n, m);
    for (int i = 0; i < n; i++)
        for (int j = 0; j < m; j++)
            scanf("%d", &allocation[i][j]);

    printf("Enter Request matrix (%d x %d):\n", n, m);
    for (int i = 0; i < n; i++)
        for (int j = 0; j < m; j++)
            scanf("%d", &request[i][j]);

    printf("Enter Available resources (%d): \n", m);
    for (int i = 0; i < m; i++)
        scanf("%d", &available[i]);

    detectDeadlock();
    return 0;
}
```

OUTPUT:

```
Enter number of processes : 2
Enter number of resources : 2
Enter Allocation matrix (2 x 2):
1 0
0 1
Enter Request matrix (2 x 2):
0 1
1 0
Enter Available ~~matrix~~ Resources(2):
0 0
Process 0 is in deadlock.
Process 1 is in deadlock.
```

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